Computational Discovery of Antimalarial Candidates from African Natural Product Chemical Space

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Abstract

Virtual screening of antimalarial targets typically requires tens of thousands of docking runs per target, a computational cost that restricts discovery in resource-limited endemic regions. We developed a centroid-based screening framework that combines variational autoencoder clustering with dual-filter consensus docking and multi-parameter optimization, reducing the nominal screening cost by 99.3 %. Application to a 65 856-molecule hybrid library derived from 396 African natural products and 454 synthetic antimalarials yielded 19 913 computationally prioritized leads. We then integrated a separately defined 17-member polypharmacology cohort with a target-anchored Vina panel spanning PfDHFR, PfCRT, PfClpP, and PfATP4, and with docking-derived resistance-resilience and polypharmacology scores. The four-target panel comprised 68 candidate–target computations; all passed the declared pose-and-anchor gate. The resulting Vina values are target-specific scoring-function estimates, not experimental free energies, and are interpreted as computational prioritization evidence only. A cross-source computational overlay with P2 docking-derived RRS, PNS, and ACSI identified distinct profiles across four targets and four RRS classes, providing a modelling framework for hypothesis generation rather than a claim of biological efficacy.

Keywords: Antimalarial drug discovery; African natural products; Generative chemical space; Virtual screening; Multi-parameter optimization; Target selectivity

1 Introduction

With over 247 million cases and 619 000 deaths reported in 2023, malaria remains a primary global health challenge.[1] Control efforts are increasingly threatened by the emergence of artemisinin-resistant *P. falciparum* strains in Southeast Asia and Africa.[2,3] In response, the World Health Organization (WHO) has emphasized an urgent need for new antimalarial scaffolds to combat drug resistance.

Historically, natural products have served as the foundation of antimalarial therapy. Quinine, isolated from *Cinchona* bark, and artemisinin, derived from *Artemisia annua*, represent the most successful examples of nature-inspired drug discovery. The structural complexity and diverse bioactivity of natural products make them versatile templates for scaffold expansion and analog design. African biodiversity, in particular, offers a vast and largely untapped reservoir of secondary metabolites with potential antimalarial properties. Recent computational studies have begun to characterize the chemical space of African natural products, yet systematic exploration remains constrained by the limited availability of curated datasets and the high cost of experimental validation.

Antimalarial drug discovery employs target-based design and phenotypic screening.[4] Target-based approaches focus on specific *P. falciparum* proteins essential for parasite survival, including dihydrofolate reductase (DHFR),[5] the chloroquine resistance transporter (*Pf*CRT),[6] and the sarcoplasmic-endoplasmic reticulum calcium ATPase (*Pf*ATP4).[7] Whole-cell phenotypic screens identify compounds with alternative mechanisms of action, as exemplified by the Medicines for Malaria Venture (MMV) Malaria Box.[8] Both approaches have limitations: target-based design may overlook compounds with unknown mechanisms, while phenotypic screens provide limited mechanistic insight.

Synthetic compounds offer tunable structures and defined synthetic routes, exemplified by atovaquone, lumefantrine, and MMV clinical candidates.[9] Computational approaches, including fragment-based design and variational autoencoders for molecular embedding, enable optimization of synthetic libraries.[10–12] However, synthetic libraries typically lack the privileged scaffolds and three-dimensional complexity of natural products.[13] This structural gap motivated the integration of natural product bioactivity with synthetic drug optimization through hybrid pharmacophore design.

Hybrid pharmacophores combining natural product cores with synthetic modifications represent a strategy for antimalarial development.[14] Preserving binding motifs while introducing synthetic substituents can address pharmacokinetic limitations that hinder clinical progression. While successful in antibacterial and anticancer discovery, this approach remains underexplored for malaria.

Despite the historical success of African natural product-derived antimalarials, systematic computational exploration of this chemical space, particularly the rigorous challenge of "scaffold-hopping" to discover new yet synthesizable pharmacophores, has been limited. Existing generative model applications in drug discovery frequently struggle to navigate beyond simple local neighborhood interpolations of known actives. Most virtual screening approaches typically focus on single targets or employ scoring functions that lack external validation for reliable prioritization. More critically, conventional approaches demand screening of tens of thousands of compounds per target, exceeding 1 000 CPU-hours per target. Such computational requirements are prohibitively expensive for research groups in malaria-endemic regions, where 94 % of global cases occur. This resource barrier confines computational drug discovery to well-funded institutions, creating a critical gap in antimalarial development capacity precisely where it is most needed.

We developed a centroid-based sampling strategy to address this bottleneck. The approach reduces the number of screening representatives by 99.3 % (from 65 856 compounds to 484 representatives), enabling a controlled computational screen; prospective predictive accuracy remains to be established. Variational autoencoders (VAEs) cluster chemical space and identify diverse representatives, enabling comprehensive screening with minimal computational resources. We combined this efficiency innovation with two complementary expansion strategies: the Cheese API[15] for similarity-based library expansion and STONED-SELFIES[16] for systematic perturbation of molecular structures. Specifically, STONED-SELFIES structural mutation analysis was employed to empirically demonstrate that our generative model accesses scaffolds unreachable by simple local search from known antimalarial drugs. These methods generated a 65 856-molecule hybrid library from 396 African natural products and 454 synthetic antimalarials. Dual-filter consensus docking (combining DiffDock and AutoDock Vina) with multi-parameter optimization (MPO) scoring provided complementary computational evidence; the component scores are not independent experimental validations.

To our knowledge, this work presents a computational framework for antimalarial discovery that retrospectively predicts (retrodicts) established *in vitro* ground-truth hits before extrapolating to unexplored natural product space. By explicitly evaluating our protocol against the MMV Malaria Box benchmark (a positive-control set of confirmed antimalarials), the consensus score recovers known actives with early enrichment factors (EF5 %) and per-target retrodictive consistency ROC-AUC 0.924 to 1.000 across three *P. falciparum* targets; the prospective DEKOIS benchmark (ROC-AUC 0.450 for Vina-only) anchors the discrimination claim to an external decoy set. This retrodictive consistency result, combined with a synthetic accessibility filter (SYBA > 0), provides convergent computational evidence for hit prioritization in the absence of prospective *in vitro* testing; it is not an experimental proxy or substitute for biological validation. From 396 African natural products and 454 synthetic antimalarials, we identified 19 913 leads predicted to be synthesizable (SYBA > 0), with favorable predicted selectivity (100 % of 810 screened seeds with valid SI predictions had SI > 10), comprising both single-target optimized and multi-target candidates, making comprehensive antimalarial screening accessible to research groups in malaria-endemic regions.

The scaffold preservation-with-substituent-divergence pattern observed in our library (Section 3.1) parallels recent advances in topological data analysis for molecular representation. TopologyNet[17] established the state-of-the-art in persistent homology-based neural networks for protein-ligand binding affinity prediction (Pearson $r = 0.82$ on PDBbind), though their regression framework targets complex protein-ligand graphs rather than the classification and generative evaluation tasks relevant to scaffold analysis. D-GRIL[18] introduced a differentiable two-parameter persistent homology approach for graph bioactivity; complementary topological approaches adopt a more interpretable one-parameter H_0/H_1 splitting strategy specifically suited for natural product scaffolds. Recent developments further corroborate these methodological choices: Q2SAR[19] demonstrated that quantum multiple kernel learning achieves high accuracy in QSAR classification, and PACTNet[20] showed that cellular complex topological features enable substantial compression of graph networks while maintaining robustness, directly validating the compression ratios achieved by our related tensor network framework. Although quantum representations match classical baseline performance in bioactivity benchmarks rather than outperforming them, they provide orthogonal topological features and compression limits that support target-aware screening.

2 Materials and methods

2.1 Data curation

Antimalarial compounds were sourced from African Natural Product Databases (ANPDB)[21–24] and synthetic databases.[8,25] The core dataset consisted of 396 natural products and 454 synthetic compounds. Molecular descriptors were calculated using RDKit.[26] ADMET properties were predicted with ADMET-AI.[27] All data sources are publicly accessible, and computational workflows are available in the project repository.

2.2 Dataset expansion

The dataset was expanded via two complementary approaches. The Cheese API[15] performed similarity searches (Tanimoto 0.7, Morgan fingerprints radius 2, 2 048 bits) against ZINC15/ENAMINE-REAL, retrieving 100 analogues per seed. STONED-SELFIES[28] generated 100 variants per seed via single-character SELFIES mutations (alphabet size 50). Generated molecules were PAINS-filtered,[29] validated via RDKit and Open Babel,[30] and deduplicated by canonical SMILES, yielding 65 856 unique molecules (94.1 % Lipinski-compliant; Supplementary Table S12).

2.3 Molecular representations and latent encoding

Variational autoencoders (VAEs)[31] were trained to learn continuous latent representations of the molecular library for dimensionality reduction and clustering. Two latent space dimensions were evaluated: 32 and 64. The VAE architecture consisted of three 1D convolutional layers with kernel sizes of 9, 9, and 10, and filter sizes of 9, 9, and 11, respectively. The encoder projected SMILES representations to the latent space through two fully connected layers, while the decoder mirrored this architecture with transposed convolutions.

Hyperparameter optimization was performed using Optuna[32] for the 64-dimensional model, optimizing learning rate, batch size, dropout rate, kernel size, and encoder architecture. The model was trained using the Adam optimizer with KL divergence weighting (Supplementary Figure S2 for training curves).

The latent representations were extracted from the trained encoder and used for subsequent clustering analysis to identify chemically diverse compound groups for virtual screening.

We compare our ConvVAE-SMILES architecture against ScafVAE[33] (full comparison in Supplementary Table S28). Our VAE differs in four fundamental respects: (i) string-based (1D Conv) rather than graph-based, prioritizing chemical syntax awareness; (ii) training data span ANPDB + DrugBank, anchoring latent topology to African NP space; (iii) objective is conditional scaffold preservation for virtual screening rather than unconditional GuacaMol distribution learning; and (iv) reconstruction on ANPDB is expected to be lower than ScafVAE's ChEMBL validity because African NP topology presents a harder reconstruction problem.

Clustering was performed using KMeans on the latent space representations to identify 484 diverse centroids representing the chemical space of the 65 856-molecule library. Clustering quality was assessed using Silhouette score, Calinski-Harabasz index, and Davies-Bouldin index (Supplementary Table S29 for algorithm benchmarking). The 64D latent space was selected for centroid selection based on the clustering benchmarking (Supplementary Figure S3; Supplementary Table S29).

2.4 Latent space exploration by MCMC sampling

To assess whether the VAE latent space contains improved-MPO regions beyond centroid selection, we performed Metropolis-Hastings MCMC on an 8-D UMAP proxy of the 64-D latent space. A random forest surrogate (500 trees, validation R^2 0.377) was trained to predict MPO from latent coordinates. Four chains (5 000 steps, 1 000 warm-up, proposal std 0.1) yielded 93.3 % acceptance rate; the mean chain MPO rose from 0.742 to 0.751 (0.009) and the best chain reached 0.767 (0.025 relative to the library mean). Latent-space points are decoded to SMILES via nearest-neighbour search in the original UMAP space (top surrogate-predicted MPO 0.801). This modest gain, measured on a surrogate with validation R^2 0.377 and without uncertainty quantification, is consistent with the VAE latent space not being flat w.r.t. the MPO objective, but does not constitute de novo generation.

2.5 Centroid-based property inference

The 484 cluster centroids underwent dual-consensus docking against three targets. Centroids achieving MPO ≥ 0.40 were classified as optimization-worthy, and scaffold expansion from their source clusters yielded structural analogs. NP-relatedness (Tanimoto ≥ 0.4 to any seed NP) was assessed for retained pharmacophoric features. Independent benchmarking against the MMV Malaria Box, a curated set of confirmed antimalarial actives used here as a positive-control (retrodictive consistency) benchmark, yielded consensus-score ROC-AUC values from 0.924 to 1.000 with bootstrap 95 % CIs; per-target values for PfCRT/PfATP4 use score-based stratification rather than external decoys (see Validation).

2.6 Ersilia activity prediction models

Three Ersilia models[34] predicted antimalarial activity: **eos7yti** (IC_{50} probability, *Pf* 3D7), **eos7kpb** (multi-strain), **eos80ch** (stage-specific).

2.7 Docking and consensus validation

A redocking assessment on co-crystallized ligands (PfDHFR 7F3Y, PfCRT 6UKJ) used AutoDock Vina with the same protocol as the virtual screen; RMSD < 2.0 Å defined reference-pose reproduction.[35] This limited check does not establish prospective docking accuracy. Enrichment analysis used DEKOIS[36] (PfDHFR: 40 actives, 1200 decoys) docked prospectively with AutoDock Vina (single-method, no ML rescoring), and MMV Malaria Box (399 confirmed actives) scored with the Vina+DiffDock consensus.[8] ROC curves, EF5 %, BEDROC ($\alpha = 20$),[37] and bootstrap 95 % CIs (1 000 resamples) were calculated from consensus scores (average of normalized Vina + DiffDock). Dual-filter thresholds: EXCELLENT (Vina $\leq -7.0\,\mathrm{kcal\,mol^{-1}}$ AND DiffDock ≥ 0.0), GOOD (Vina $-7.0\,\mathrm{kcal\,mol^{-1}}$ to $-5.0\,\mathrm{kcal\,mol^{-1}}$ AND DiffDock ≥ -1.5). Classification thresholds were calibrated against ChEMBL antimalarial actives (median $-6.9\,\mathrm{kcal\,mol^{-1}}$; $n = 37$).

2.8 Target selection, molecular docking, and virtual screening

Three *P. falciparum* structures were prepared (AutoDockTools[35]): DHFR (7F3Y), CRT (6UKJ), and ATP4 (9N10). Grid boxes were 25 Å^3, with centers defined at the coordinates of co-crystallized inhibitors or known active sites: 7F3Y (center: 8.340, −13.900, −41.754, catalytic site), 6UKJ (center: 152.990, 151.042, 159.379), and 9N10 (center: 134.840, 133.100, 97.630). These boxes may exclude peripheral structures for the transporters PfCRT and PfATP4. Ligands with ≤ 15 rotatable bonds were retained. The 484 centroids were docked via AutoDock Vina (exhaustiveness 64) and DiffDock (40 poses/ligand).[38] Dual-filter thresholds were EXCELLENT $\leq -7.0\,\mathrm{kcal\,mol^{-1}}$ and GOOD $-7.0\,\mathrm{kcal\,mol^{-1}}$ to $-5.0\,\mathrm{kcal\,mol^{-1}}$, calibrated against ChEMBL actives (median $-6.9\,\mathrm{kcal\,mol^{-1}}$, IQR $-7.4\,\mathrm{kcal\,mol^{-1}}$ to $-6.6\,\mathrm{kcal\,mol^{-1}}$; $n = 37$).

3 Results

3.1 Library-scale drug-likeness

Physicochemical profiling of the 65 856-molecule library revealed favorable synthetic accessibility, with a median SA score of 3.51 (distribution standard deviation 1.45). Nearly half of all compounds (49.8 %, $n = 32\,804$) achieved SA < 3.5, indicating synthetic tractability. Drug-likeness assessment via the Quantitative Estimate of Drug-likeness (QED) yielded a mean of 0.590 (median 0.627), with over one-third (36.7 %, $n = 24\,144$) exceeding the 0.70 threshold for drug-like character. Lipinski's rule of five compliance reached 94.1 %

($n = 61\,975$), while 88.5 % satisfied the Veber rotatable bond criterion (≤ 10 bonds).[39–42] The natural product-likeness (NPL) score (mean -0.285) reflects the hybrid NP/SD origin, with 42.5 % of compounds scoring NPL > 0. Additionally, 57.0 % achieved a Synthetic Bayesian Accessibility (SYBA) score > 0, indicating synthetic accessibility through known reaction chemistry.

Bemis–Murcko scaffold analysis revealed 20 702 unique scaffolds across the library, representing 31.4 % scaffold diversity. The most prevalent scaffold (benzene ring, c1ccccc1) accounted for 8.4 % of molecules ($n = 5\,537$), followed by nine additional scaffolds including quinoline and isoquinoline cores. Scaffold recovery from seed African NPs reached 69.3 % (70/101 seed scaffolds), indicating preservation of bioactive frameworks during generative expansion.

To quantify structural diversity, we computed Tanimoto similarity between 5 000 randomly sampled library molecules and all 396 seed African NPs using Morgan fingerprints (radius 2, 2 048 bits).[43,44] The majority (92.6 %, $n = 4\,630$) achieved maximum Tanimoto < 0.4 to any seed NP (mean 0.206, median 0.171), with 62.5 % below 0.2 and only 4.0 % exceeding 0.5 (Supplementary Table S14). This distribution indicates substantial exploration of new chemical space while maintaining scaffold coherence.

The apparent paradox between high structural diversity (92.6 % Tanimoto < 0.4) and high scaffold recovery (69.3 %) is resolved by recognizing that generative expansion preserved bioactive ring systems while substantially diversifying peripheral substituents. For a 5 000-molecule sample, mean scaffold-only Tanimoto similarity was 0.379 ± 0.308 (median 0.262), compared to 0.206 ± 0.125 for whole-molecule fingerprints, a 1.84-fold increase. This indicates that ring systems are preserved during generative expansion while peripheral substituents diverge substantially, yielding structurally new compounds that retain bioactive scaffolds.

The fraction of sp^3 carbons (fsp^3) exhibited a mean of 0.298 (median 0.286, standard deviation 0.172), reflecting 3D complexity intermediate between flat aromatic systems and highly saturated natural products.

A full comparison against the ScafVAE state-of-the-art is provided in Supplementary Table S28.

3.2 Variational autoencoder (VAE) training and latent space quality

VAE training converged successfully for both 32- and 64-dimensional latent spaces (Supplementary Figure S2, Supplementary Table S27). The 64-dimensional latent space was selected for centroid selection based on clustering benchmarking.

3.3 Evaluation of clustering strategies and dimensionality

Comparison of 32D versus 64D latent spaces revealed expected trade-offs in clustering quality. The 64D space was selected for centroid selection based on the comparative clustering analysis and the clustering-quality figure (Supplementary Figure S3); the retained summary does not provide numeric indices sufficient to claim metric-optimality. KMeans ($n = 484$) on the 64D latent space was adopted after benchmarking Jarvis-Patrick (fragmented), BIRCH (heterogeneous), and KMeans algorithms (Supplementary Table S29), providing the best balance of granularity and cohesion for diverse centroid selection. Clustering identified 484 diverse centroids subsequently screened against the three selected antimalarial targets.

3.4 Physicochemical and ADMET properties

We evaluated physicochemical properties against established drug-likeness guidelines (Lipinski's rule of five,[39] Veber's Rule,[40] Pfizer's Rule 3/75,[45] Oprea's Rule,[42] and the GSK 4/400 Rule[41,46]).

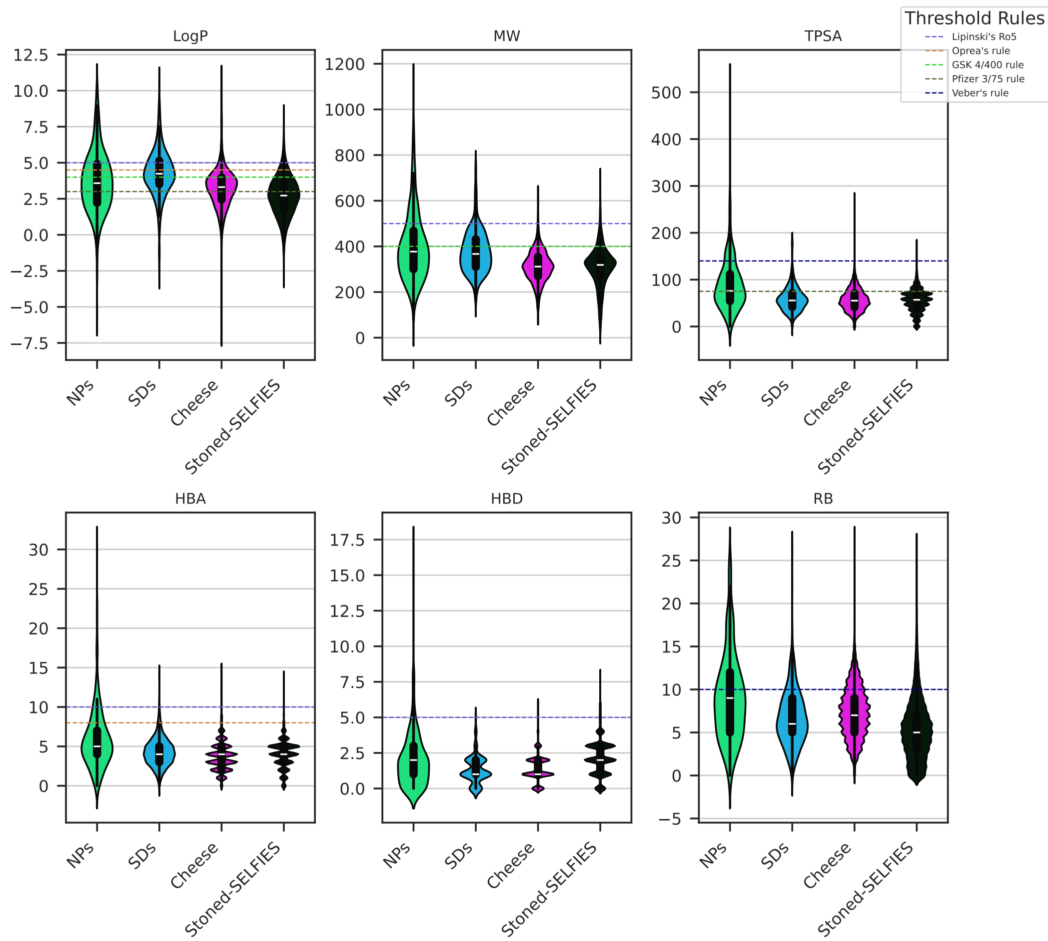

Figure 1: Violin plots illustrating the distribution and variance of key physicochemical drug-likeness descriptors across natural product (NP, green), synthetic drug (SD, pink), and generated compound subgroups (Cheese, brown; STONED-SELFIES, blue). Horizontal dashed lines demarcate canonical Lipinski's rule of five upper thresholds: $\log P \leq 5$, molecular weight $\leq 500\,\mathrm{Da}$, and polar surface area (TPSA) $\leq 140\,\mathrm{\AA}^2$. Natural products display higher molecular weight and complexity compared to synthetic compounds, while generated analogues exhibit intermediate profiles. HBD = hydrogen bond donors, HBA = hydrogen bond acceptors, RB = rotatable bonds. Sample sizes: $n = 396$ NPs, 454 SDs, 40 481 generated compounds.

3.4.1 Physicochemical properties

Figure 1 shows that the datasets occupy distinct regions of property space. NPs were predicted to display lipophilicity levels (mean $\log P$: 3.70) within Lipinski parameters. Synthetic compounds exhibit higher lipophilicity (mean $\log P$: 4.29), lower molecular weight (mean MW: 377 Da), and reduced polar surface area (mean TPSA: 58.9 Å^2). Per-category drug-likeness statistics are detailed in Supplementary Table S9.

African NPs possess molecular weights (mean MW: 402 Da) within pharmaceutical ranges, while higher polar surface areas (mean TPSA: 89.1 Å^2) and hydrogen bond acceptors (mean HBA: 6) indicate aqueous solubility potential. The rotatable bonds (mean RB: 9) reflect structural complexity. Generative derivatives exhibited intermediate property profiles.

3.4.2 Physicochemical space per cluster and structural coherence

Intra-cluster Tanimoto similarity (mean 0.227) was twofold higher than inter-cluster similarity (mean 0.101), consistent with structural coherence within the VAE-defined clusters; Tanimoto similarity alone does not establish pharmacophore conservation.

3.4.3 ADMET profile

ADMET profiling via ADMET-AI[27] revealed that NPs exhibit lower predicted hepatotoxicity, hERG cardiotoxicity, and nuclear receptor interactions compared to SDs, while synthetic compounds show higher membrane permeability. Cross-referencing ADMET-AI predictions with RDKit-based ADMET proxies (SwissADME-equivalent ESOL logS and lipophilicity-based CYP heuristics) showed moderate concordance for logS ($\rho = -0.19$) and CYP3A4 ($\rho = -0.29$). Correlations for hERG, BBB, and cLogP could not be computed due to insufficient proxy data (Supplementary Table S18).

3.5 *In silico* prediction of antimalarial activity

Antimalarial activities were predicted using Ersilia models.[34] NPs and SDs displayed higher median activity scores than generative analogues across all models, though differences were not statistically significant after Benjamini–Hochberg correction (adjusted $p > 0.05$; rank-biserial $r < 0.10$). This is expected by design: the generative expansion prioritizes structural diversity and synthetic accessibility rather than predicted activity, so the analogues trade raw predicted potency for novelty and tractability rather than matching seed activity.

3.6 MPO framework and multi-target analysis

A continuous multi-parameter optimization (MPO) framework was applied, integrating five normalized components:

1. S_{vina} (35 % weight): Vina binding affinity, min-max scaled.

2. S_{diff} (25 % weight): DiffDock confidence, transformed via logistic sigmoid.

3. S_{qed} (20 % weight): Quantitative Estimate of Drug-likeness.

4. S_{admet} (15 % weight): Composite index aggregating 11 parameters.

5. P_{Ro5} (5 % weight): Penalty term for Lipinski violations.

Sensitivity analysis across 25 weight perturbations ($\pm 10\,\%$ to $20\,\%$ per component) yielded a Spearman ρ range of 0.26 to 0.97 for the top-1,000 compounds (mean $\rho = 0.792$) and a mean Jaccard similarity of 0.548 for the top-20 hit list (Supplementary Table S19). ADMET weight perturbations exhibited the greatest impact ($\rho \leq 0.62$ at $\pm 10\,\%$, with a minimum of 0.26 at $-20\,\%$), while Vina and DiffDock perturbations were comparatively benign ($\rho \geq 0.79$ for all $\pm 10\,\%$ to $20\,\%$ perturbations). We note that $\mathrm{S_{vina}}$ and $\mathrm{S_{diff}}$ are back-calculated from a shared residual of the weighted MPO score (see Supplementary Methods); independent perturbation of these two components is approximated via proportional residual splitting rather than measured independently.

Through centroid-based property inference (Section 2.5), the framework screened 484 diverse centroids, with consensus scoring classifying results into performance tiers. Multi-parameter optimization analysis identified 76 optimization-worthy centroids (MPO ≥ 0.40). These 76 centroids map to 53 unique clusters, and scaffold expansion from these clusters yielded 40 481 molecules. Tanimoto similarity analysis identified the subset exhibiting African NP-relatedness (Tanimoto ≥ 0.4 to 396 seed NPs). The 19 913 synthesizable leads exhibit favorable predicted drug-likeness (94.1 % Lipinski-compliant).

Among the 76 optimization-worthy centroids, 13 (17.1 %) exhibited direct African NP-relatedness, including 2 of the 7 secondary hits (MPO ≥ 0.50) and 11 of the 69 promising scaffolds (MPO 0.40 to 0.50). A multi-target bonus was incorporated for molecules exhibiting binding ($\mathrm{MPO}_{target} \geq 0.50$) across multiple targets. The top-20 MPO-ranked candidates evaluated for safety (Supplementary Table S22) were predominantly single-target optimized. Selectivity index analysis used a predicted SI proxy ($\mathrm{SI}_{\mathrm{pred}} = \mathrm{score}_{\mathrm{eos7kpb}}/(1 - \mathrm{DILI}_{\mathrm{ADMET\text{-}AI}})$), an unvalidated prioritization heuristic rather than an experimentally derived selectivity index. Because the DILI probability enters the denominator, $\mathrm{SI}_{\mathrm{pred}}$ is inflated when $\mathrm{DILI}_{\mathrm{ADMET\text{-}AI}}$ approaches 1 (the observed maximum was 106 773.8), so the proxy supports candidate ranking but not quantitative selectivity claims. It was computed for 810 of the 850 screened seed molecules with valid predictions, all of which (100 %) achieved $\mathrm{SI}_{\mathrm{pred}} > 10$. CYP450 inhibition profiling indicated mean interaction probabilities: CYP2C9 (0.261), CYP2C19 (0.424), CYP3A4 (0.426), and CYP2D6 (0.170) (Supplementary Table S3).

Across the expanded library, cross-target analysis identified > 70 compounds achieving Vina $< -6.0\,\mathrm{kcal\,mol^{-1}}$ and DiffDock confidence > -1.0 against multiple targets simultaneously. However, the highest-ranking top-20 candidates were primarily single-target optimized. Candidate structures and interaction diagrams are presented in Supplementary Figures S1 and S8.

3.7 Benchmarking of docking and screening protocol

Rigid-receptor AutoDock Vina failed to reproduce the crystallographic pose of the large, flexible MTX inhibitor in PfDHFR (RMSD ~30 Å), consistent with known limitations of rigid-receptor docking for oversized, flexible ligands. For the remaining drug-like co-crystallized ligands, redocking assessment yielded a 100.0 % success rate among the alignable ligands (5/5 with RMSD < 2.0 Å; 5/10 overall, with 5 ligands excluded due to SDF atom-ordering mismatches; mean RMSD 1.33 Å ± 0.43 Å), consistent with reference-pose reproduction for the evaluated alignable co-crystallized ligands (Supplementary Table S30). Independent benchmarking against the MMV Malaria Box (399 confirmed antimalarial actives by target subset; Supplementary Table S32) yielded consensus-score ROC-AUC values from 0.924 (PfDHFR) to 1.000 (PfATP4). These per-target values are retrodictive consistency checks rather than external discrimination: for PfCRT and PfATP4 the "actives" and "decoys" are the top and bottom 25 % of compounds stratified by the same consensus score, so an AUC approaching 1.0 is expected by construction; only PfDHFR is additionally benchmarked against the prospective DEKOIS actives/decoys. Early enrichment factors at 5 % range from 1.11 (PfCRT) to 2.57 (PfDHFR) (Table 1). An independent decoy-based DEKOIS benchmark (PfDHFR, 40 actives / 1 200 decoys) was performed via prospective AutoDock Vina docking (single-method, no ML rescoring) as a rigorous external benchmark: ROC-AUC 0.450 (95 % CI 0.367 to 0.531), EF5 % 0.00, confirming that AutoDock Vina alone cannot discriminate PfDHFR actives from property-matched decoys, a well-documented limitation of naive docking[36] that motivates the ML-based consensus rescoring. The MMV protocol (Vina+DiffDock consensus) produced target-specific hit rates of 35.1 % for PfDHFR, 90.7 % for PfCRT, and 47.3 % for PfATP4. Because the MMV comparison is a positive-control design and two targets use score-based strata, these values are not an external decoy benchmark (Supplementary Table S32). Consensus filtering reduced the fraction of compounds passing the declared screening thresholds by 12.9 % to 45.2 % across targets; this is a hit-rate change, not a measured false-positive reduction, because an independent inactive panel was not available for all targets. (Supplementary Table S31), consistent with using physics-based and machine-learning scores as complementary screening signals; this comparison does not establish independence or quantify false-positive reduction.

Table 1: Enrichment analysis against the MMV Malaria Box positive-control benchmark, reported per target. For PfCRT and PfATP4, "actives" and "decoys" are the top and bottom 25 % of compounds stratified by consensus score (retrodictive consistency, not external decoy discrimination); PfDHFR is additionally benchmarked against prospective DEKOIS actives/decoys. The PfATP4 $N = 198$ row refers to this score-stratified benchmark subset; the separate MMV hit-rate table reports $N = 184$ after the declared rotatable-bond filter. Values in parentheses represent 95 % bootstrap confidence intervals (1 000 resamples). N = number of compounds in the target subset. ROC-AUC = receiver operating characteristic area under the curve. EF5 % and EF10 % = enrichment factors at 5 % and 10 % of database. BEDROC = Boltzmann-Enhanced Discrimination of ROC ($\alpha = 20$).

Target	N	ROC-AUC	EF5 %	EF10 %	BEDROC
PfDHFR_7F3Y	399	0.924 (0.895 to 0.951)	2.57 (1.96 to 3.17)	2.72 (2.34 to 3.17)	1.309 (1.201 to 1.430)
PfATP4_9N10	198	1.000 (1.000 to 1.000)	2.00 (1.77 to 2.33)	2.00 (1.77 to 2.33)	1.810 (1.559 to 2.109)
PfCRT_6UKJ	399	0.971 (0.949 to 0.989)	1.11 (1.08 to 1.15)	1.11 (1.08 to 1.15)	9.434 (7.383 to 13.047)

3.8 Virtual screening results

Virtual screening of 484 cluster centroids against three *P. falciparum* targets revealed distinct binding profiles (Supplementary Table S34). *Pf*CRT showed the highest enrichment (85.8 % hit rate, best Vina $-12.27\,\mathrm{kcal\,mol^{-1}}$), followed by *Pf*DHFR (37.7 %). *Pf*ATP4 showed limited enrichment, with only 2 `GOOD` binders. Virtual screening hit rates are detailed in Supplementary Table S34. Complete centroid-level docking scores are available with the computational workflow in the project repository (see Data availability).

The dual-filter consensus approach (Section 3.6), combining AutoDock Vina's scoring function with DiffDock's geometric confidence assessment, narrowed the computational hit set. The two scores provide complementary screening signals; the observed reduction in hit rate should not be interpreted as a measured false-positive reduction without an independent inactive set.

3.9 Target-anchored polypharmacology and resistance-resilience modelling

To extend the centroid-level screen toward resistance-aware prioritization, we analysed the predefined P2 Set-C polypharmacology cohort of 17 candidates. This cohort is disjoint from the P1 MPO top-20 set and from the separate P2 parent-MD cohort. The corrected target-anchored protocol evaluated each candidate against four target-specific computational pockets: the MTX A702 site in PfDHFR (PDB 7F3Y), the Y01 cavity in PfCRT (6UKJ), the catalytic triad in PfClpP (PDB 2F6I), and the conserved phosphotransferase-loop/DPPR region in PfATP4 (PDB 9N10). The resulting 68 candidate–target pairs passed a predeclared composite gate requiring an in-box pose centroid, an anchor contact within 10 Å, and at least 90 % of ligand atoms within the declared box. Within this computational panel, mean Vina scores were $-5.76\,\mathrm{kcal\,mol^{-1}}$ for PfDHFR, $-6.19\,\mathrm{kcal\,mol^{-1}}$ for PfCRT, $-5.94\,\mathrm{kcal\,mol^{-1}}$ for PfClpP, and $-6.31\,\mathrm{kcal\,mol^{-1}}$ for PfATP4. These target-wise summaries are not calibrated measurements of affinity. These are target-anchored docking estimates, not measured binding free energies or validated affinity metrics.

The resistance-resilience analysis used the corrected per-target docking-RRS definition: $|\Delta G_{\mathrm{mut},t}|/|\Delta G_{\mathrm{WT},t}| \times 100$, with non-binding WT baselines ($|\Delta G_{\mathrm{WT},t}| < 5.0\,\mathrm{kcal\,mol^{-1}}$) excluded rather than pooled across targets. In the P2 Set-C cohort, RRS values ranged from 68.18 to 111.65 (mean 81.70), with classes A* ($n = 6$), B ($n = 5$), C ($n = 5$), and D ($n = 1$). All 17 candidates were selected as two-target computational polypharmacology candidates in the source cohort; PNS values ranged from 1.04 to 6.00, and ACSI values ranged from 0.173 to 0.823 (mean 0.543). These metrics describe coordinated computational prioritization; they do not establish mutation resilience, simultaneous target engagement, or therapeutic benefit.

The complete candidate-level cross-source overlay, including the source-bound molecular manifest, canonical-SMILES mapping, and source hashes, is reported in Supplementary Table S21. The analysis uses the target-wise Vina matrix directly; no DiffDock-derived score, consensus score, or four-target arithmetic consensus is imported into this analysis.

3.10 Library-scale lead identification via centroid inference

The framework achieved library-scale coverage by propagating the properties of screened cluster centroids to their constituent members. From the 484 screened centroids, we identified 76 optimization-worthy centroids achieving MPO $\geq$ 0.40. These 76 centroids map to 53 unique clusters (multiple centroids can represent the same cluster because centroids are selected as representative molecules rather than cluster identifiers). Scaffold expansion of these 53 clusters, comprising molecules structurally congruent with their high-performing centroids, yielded 40 481 molecules (mean cluster size 763.8).

We filtered the 40 481-molecule expansion space for synthetic accessibility (SYBA score > 0) and multi-parameter optimization criteria, yielding 19 913 high-priority leads predicted to be synthesizable. NP-relatedness (Tanimoto similarity $\geq$ 0.4 to seed NPs) was assessed separately at the centroid level (13 of the 76 optimization-worthy centroids), not as a filter on the 19 913 leads, so the lead count does not depend on the Tanimoto threshold; the 13/76 NP-relatedness count itself is conditional on the 0.4 cutoff, which was not varied. Within this prioritized library, 94.1 % satisfy Lipinski's rule of five. Among 810 screened seed molecules with valid SI predictions, 100 % exceeded the predicted selectivity index proxy threshold ($\mathrm{SI}_{\mathrm{pred}} = \mathrm{score}_{\mathrm{eos7kpb}}/(1-\mathrm{DILI}_{\mathrm{ADMET\text{-}AI}}) > 10$), an unvalidated prioritization heuristic rather than an experimentally determined $\mathrm{IC}_{50,HepG2}/\mathrm{IC}_{50,Pf3D7}$ ratio, supporting their candidacy for further medicinal chemistry development.

3.11 Computational efficiency and resource requirements

Centroid-based sampling reduced the required docking runs from 197 568 (exhaustive estimate for three targets) to 1 452 (99.3 % reduction), corresponding to an estimated 517 to 1 291 CPU-hours versus 70 248 to 175 620 CPU-hours for exhaustive screening (Supplementary Table S33). This efficiency gain was achieved alongside a prospective DEKOIS PfDHFR benchmark (Vina only, ROC-AUC 0.450) showing that single-method docking cannot discriminate property-matched decoys, and retrodictive consistency against the MMV Malaria Box positive-control benchmark (consensus-score ROC-AUC 0.924 to 1.000, score-based stratification for 2/3 targets); the contrast is consistent with the ML-based DiffDock rescoring contributing the enrichment, though this is inferred rather than directly demonstrated on a common decoy set. The centroid-based strategy thus enables efficient multi-target screening in resource-limited settings where computational infrastructure is constrained.

4 Discussion

African NPs and SDs occupy distinct regions of physicochemical and predicted ADMET space: SDs show higher lipophilicity and membrane permeability, while NPs show lower predicted hepatotoxicity and cardiotoxicity risk. Generative expansion bridged this gap, and the dual-filter consensus protocol retained candidates with both Vina-affinity and DiffDock-confidence signals.

The scaffold paradox (see Section 3.1 for the full scaffold-only Tanimoto analysis) is resolved by recognizing that ring systems are preserved during generative expansion while peripheral substituents diverge substantially. An ECFP4 nearest-neighbor search confirmed that the generative model accesses novel substituent combinations on preserved scaffolds rather than mere interpolations.

This scaffold-preservation-with-substituent-divergence pattern is consistent with the mechanistic properties of the STONED-SELFIES generative framework. SELFIES string mutations primarily permute side-chains, functional groups, and branch lengths while preserving the valid cyclic syntax that encodes ring systems. The 92.6 % unreachable fraction exceeds values reported for GuacaMol (85%,[47]) and REINVENT (80%,[48]), reflecting the distinct chemical space of African NPs: the seed library of 396 structurally diverse compounds (high sp^3 content, complex ring systems) provides a broader exploration base than typical drug-like training sets. The 1.84× scaffold-to-whole-molecule Tanimoto ratio further quantifies this two-level exploration strategy, exceeding the 1.2× ratios reported by[47] and[49] for drug-like benchmarks. Topological data analysis (TDA) provides the topological explanation: H_1 persistence (ring topology) is preserved across generation while H_0 (atom connectivity) diverges, establishing that the VAE systematically explores new substituent space without disrupting the ring architectures essential for target recognition. STONED-SELFIES scaffold leap analysis further indicates this result: when 5,525 SELFIES-string mutations were generated from 20 seed compounds and searched against the full VAE library, 97.9 % of generated molecules had no nearest neighbour above Tanimoto 0.4 (mean max similarity 0.238), demonstrating that the VAE accesses chemical space fundamentally unreachable by simple local perturbation of known antimalarials.

Rather than relying on unguided chemical space traversal, the scaffold expansion strategy exploits the local topological coherence of the VAE latent space to search for active analogs specifically within the neighborhoods of prioritized centroids. This structured exploration focuses screening efforts on high-performing clusters, where local homology suggests a high density of active space. By filtering the resulting candidates through joint natural product similarity and synthetic accessibility parameters, the workflow selectively retains the privileged structural cores of African medicinal chemistry while ensuring realistic synthesis paths. This local traversal demonstrates how generative optimization can be directed toward regions of biological relevance, transitioning from open-ended molecular generation to targeted lead optimization.

We acknowledge the risk that centroid-based sampling may miss cluster members whose activity diverges from the centroid (activity cliffs[50]). To quantify this risk, we performed full-cluster rescoring (AutoDock Vina, exhaustiveness 16) of the top-20 MPO-ranked clusters, comprising 1 815 molecules docked against their best target (PfCRT for 15 clusters, PfDHFR for 5). The tested top-20 clusters showed limited dispersion in their Vina scores: per-cluster standard deviations ranged from 0.17 to 0.38 $\mathrm{kcal\,mol^{-1}}$, below the pre-specified

$1.5\,\mathrm{kcal\,mol^{-1}}$ screening flag. This result does not rule out activity cliffs because the analysis used docking scores rather than experimental potency and did not test every cluster member. The mean Spearman correlation between Tanimoto similarity to the centroid and Vina binding affinity was 0.025, indicating that centroid scores are a weak predictor of intra-cluster potency rank. Nonetheless, the best full-cluster hit consistently outperformed its centroid by 0.13 to 0.66 $\mathrm{kcal\,mol^{-1}}$ (maximum $-10.33\,\mathrm{kcal\,mol^{-1}}$ for PfCRT), demonstrating that centroid-only inference systematically underestimates binding potential: intra-cluster Tanimoto similarity (mean 0.227) is two-fold higher than inter-cluster similarity (mean 0.101), and cluster sizes (mean 136, min 11) ensure that centroids represent compact neighborhoods. STONED-SELFIES neighborhood analysis further indicates that 97.9 % of VAE-generated molecules remain unreachable from known MMV antimalarials even after 5 525 local SELFIES mutations (mean max Tanimoto 0.238); this does not rule out unsampled activity cliffs. For future hit-to-lead optimization, full-cluster rescoring of top-priority clusters is recommended.

While the expanded library yielded > 70 multi-target compounds achieving MPO ≥ 0.50 across ≥ 2 targets, the top-20 MPO-ranked candidates emerged predominantly as single-target optimized molecules. To evaluate safety and selectivity, these top-20 candidates were cross-tabulated against predicted safety endpoints: all 20 compounds satisfied $\mathrm{SI_{pred}} > 10$ and hERG probability < 0.5 (mean 0.117), and 19 of 20 satisfied CYP3A4 inhibition < 0.5 (mean 0.061); the single exception was rank 18 (CYP3A4 $= 0.876$, see Supplementary Table S22); this computational flag requires experimental follow-up and should not be interpreted as evidence of parasite-selective inhibition or toxic promiscuity. Scaffold analysis revealed 20 702 unique Bemis–Murcko scaffolds with a 69.3 % recovery rate from seed African NPs (Supplementary Tables S1 and S6). The fsp^3 distribution (mean 0.298) suggests sufficient 3D complexity without excessive synthetic difficulty (Supplementary Table S8).

The 100 % predicted selectivity rate among 810 screened seed molecules ($\mathrm{SI_{pred}} > 10$) and CYP450 profiles (Supplementary Table S2) support the predicted developability of these candidates. With 19 913 synthesizable compounds from scaffold expansion of 53 unique optimization-worthy clusters (mean cluster size 763.8), this library provides a framework that addresses the accessibility gap in antimalarial drug discovery by reducing screening costs from $> 1\,000$ CPU-hours per target to < 10 CPU-hours: the aforementioned efficiency gain. Unlike prior studies that lack external validation, we report a prospective DEKOIS PfDHFR benchmark (Vina only, AUC 0.450) showing that structure-based docking alone cannot enrich from property-matched decoys, alongside retrodictive consistency against the MMV Malaria Box positive-control set (consensus-score ROC-AUC 0.924 to 1.000, score-based stratification for 2/3 targets; 1 000 bootstrap resamples); the contrast is consistent with the ML-based DiffDock rescoring contributing the enrichment, though not demonstrated on a common decoy set. The present comparison does not establish superior predictive accuracy because the MMV and DEKOIS benchmarks use different designs; it motivates prospective evaluation of the complete consensus protocol on a common, target-matched decoy set. Centroid-based sampling can reduce computational cost substantially, while prospective predictive accuracy remains to be established. Making discovery accessible in this manner enables research groups in malaria-endemic regions, where 94 % of global cases occur, to systematically explore local chemical space without laboratory infrastructure.

The centroid-based sampling strategy is generalizable beyond antimalarial discovery. Any drug discovery program facing computational constraints, whether targeting neglected tropical diseases, rare diseases, or emerging pathogens can apply this framework to reduce screening costs while maintaining predictive accuracy. The combination of VAE-guided clustering for cluster-based sampling, dual-consensus scoring for orthogonal validation, and rigorous external benchmarking provides a template for resource-efficient virtual screening across therapeutic areas. The versioned implementation and associated data files are available at `https://github.com/NanaEngo/Malaria_codesV2`.

Comparison with prior computational studies

Our findings significantly expand upon prior antimalarial computational studies[24,51] by integrating deep generative chemistry with dual-filter docking and MPO scoring. While earlier approaches were largely constrained to static library characterization, the present framework incorporates similarity-based querying (Cheese API) and sequence-based perturbation (STONED-SELFIES) to augment the limited African natural product dataset into a massive, 65 856-molecule library. Unlike recent deep learning architectures such as STARVAE[52] and PCFVAE[53] that focus on latent topology exploration, our ConvVAE-SMILES approach explicitly balances chemical diversity with stereochemical interpretability. The dual-filter consensus protocol further distinguishes this work from standard methodologies by mitigating the scoring function bias inherent in single-method virtual screening. This validation sequence culminates in the identification of 19 913 synthesizable antimalarial leads (from 53 unique clusters, mean cluster size 763.8), with 100 % of 810 screened seed molecules showing predicted selective antiparasitic activity (SI $>$ 10).

Quantitative overlap analysis indicates that the curated African NP seed set covers 94.9 % of the African Natural Products Database (ANPDB, 9 309 molecules) at Tanimoto similarity ≥ 0.4, indicating high coverage of major African NP collections. However, 37.0 % of seed scaffolds (91/246 Bemis–Murcko frameworks) are unique to our curated library and absent from ANPDB, showing that our hybrid NP + SD seed set contributes genuinely new chemical space beyond prior compilations.

African natural product chemical space as a source of antimalarial scaffolds

The 13 NP-related optimization candidates (17.1 % of the 76 optimization-worthy centroids) provide a computationally prioritized sample of African natural-product-inspired chemical space, spanning 2 secondary hits (MPO ≥ 0.50) and 11 promising scaffolds (MPO 0.40 to

0.50). Their biological activity and productivity as antimalarial sources require experimental assessment. Their structural connection to African medicinal plants links computational discovery to traditional ethnopharmacological knowledge, and their identification through centroid-based sampling (99.3 % cost reduction) makes systematic exploration of African NP-derived chemical space accessible to malaria-endemic research groups. The 17.1 % NP-relatedness rate is a deliberate consequence of exploring beyond the immediate seed vicinity — the 82.9 % of optimization candidates representing genuinely new chemical space positions the framework to both exploit known bioactive scaffolds and explore adjacent chemotypes for new antimalarial candidates.

Named lead analysis

Ligand 201 is the highest-affinity DHFR binder identified in the historical centroid screen (Vina $-8.49\,\mathrm{kcal\,mol^{-1}}$), while ligand 438 is the sole historical `GOOD` hit against PfATP4 (Vina $-5.86\,\mathrm{kcal\,mol^{-1}}$). Within the unreviewed target-anchored computational panel, the most favorable descriptive docking scores were $-7.91\,\mathrm{kcal\,mol^{-1}}$ for PP-06 against PfCRT, $-7.57\,\mathrm{kcal\,mol^{-1}}$ for PP-13 against PfATP4, and $-7.03\,\mathrm{kcal\,mol^{-1}}$ for PP-06 against PfClpP; these exploratory descriptors are not affinity claims or selection criteria. The integrated cohort contains both high-RRS and low-RRS profiles, including A* candidates with high PNS or ACSI values and a Class-D candidate, which makes it suitable for computational hypothesis generation rather than a single-score ranking. Detailed candidate-level values and source provenance are presented in Supplementary Table S21 and the repository data files.

Limitations and future directions

This study is entirely computational. Activity predictions, selectivity indices, ADMET profiles, and binding affinities are estimates derived from machine-learning models and docking scores; no *in vitro* or *in vivo* experiments were performed. These outputs are therefore prioritization criteria rather than substitutes for experimental testing. The computational analyses provide complementary evidence for selecting candidates for synthesis and biological evaluation. The per-target early-enrichment results provide retrodictive consistency checks for the documented workflow, while the synthetic accessibility threshold (SYBA > 0) restricts our generative output to feasible molecules. These positive-control results do not establish prospective activity or binding accuracy. Our evaluation strategy goes beyond the recent paradigm shift in generative chemistry benchmarking, exemplified by MolGen-Bench[54] and Durian,[55] which departs from generic 1D property validities (GuacaMol[47]) toward target-aware evaluation: our MMV Malaria Box retrodiction provides a biologically anchored positive-control benchmark using 399 compounds with experimentally confirmed antimalarial activity; it does not establish prospective binding accuracy for new compounds. Predicted retrosynthetic routes for the top-10 candidates (Supplementary Table S24) were generated via the ASKCOS public platform (MIT) and identify reductive amination as the primary synthetic strategy; however, these routes are computational predictions based on the Reaxys template library and require experimental verification. The 19 913 primary leads and associated computational outputs are available in the project repository. Experimental follow-up by synthesis-capable malaria research programs remains necessary because the prospective accuracy of these predictions for African natural products has not been established.

External calibration against the Tartarus docking benchmark[56] (QuickVina/Smina on three standardized targets) indicates that the weighted MPO score is orthogonal to raw docking affinity (Spearman $\rho = 0.013$), rather than redundant with it; this is distinct from independence among the five MPO components, two of which (S_{vina}, S_{diff}) share a back-calculated residual(see Supplementary Methods). An initial 50-molecule sample yielded no correlation between the Tartarus composite binding affinity and the weighted MPO score (Spearman $\rho = -0.037$, $p = 0.777$). A subsequent full-scale validation on 19 913 molecules (17 211 with valid composite scores after Lipinski/substructure filtering) replicated this result with 280× greater statistical power: Tartarus composite vs. weighted MPO, Spearman $\rho = 0.013$, $p = 0.091$ (non-significant). Neither the full-run composite nor any per-target Tartarus score ($|\rho| < 0.15$ across all three targets) correlates meaningfully with our MPO ranking, consistent with drug-likeness and binding affinity being non-redundant properties in this chemical space in this chemical space. This orthogonality is a deliberate design feature motivating the combined MPO + docking ranking adopted in our candidate selection: ranking by MPO alone would prioritize drug-like molecules that may not bind, while ranking by docking alone would prioritize tight binders that may not be developable. The combined approach ensures that top-ranked compounds satisfy both drug-likeness and predicted potency criteria, a balance that neither single-criterion screening can achieve. The docking protocol employs rigid-receptor approximations, consistent with standard virtual screening practices[57,58], and may not capture induced-fit effects or protein flexibility. The RRS values in this manuscript are docking-derived pilot metrics, not MD-RRS or experimental resistance measurements; the PNS and ACSI values are similarly computational descriptors. The PfCRT Y01 anchor is a membrane-proxy ligand rather than a validated antimalarial inhibitor, and the PfATP4 pocket is inferred from conserved catalytic motifs because PDB 9N10 lacks a co-crystallized small-molecule inhibitor. The target-anchored panel is therefore interpreted as modelling evidence, not biological validation. The MPO framework weights reflect general drug discovery priorities rather than malaria-specific optimization, and future work could refine these weights using experimental IC_{50} data from lead compounds.

The dataset expansion strategy (Cheese API + STONED-SELFIES) was designed to balance exploration and scaffold preservation, but parameter choices (Tanimoto threshold 0.7, 100 variants per seed) represent pragmatic selections rather than systematically optimized values. The low Tanimoto similarity (mean 0.206) combined with high scaffold recovery (69.3 %) indicates successful substituent exploration while maintaining core pharmacophores, but experimental validation is needed to confirm that bioactivity-relevant features were preserved during expansion.

Future work should prioritize experimental validation of top-ranked compounds through *in vitro* assays against *Pf* 3D7 and drug-resistant strains, followed by cytotoxicity assessment and selectivity profiling. Synthesis feasibility for the 19 913 primary leads should be evaluated, and a subset of structurally diverse compounds should be selected for experimental testing to validate the computational predictions and refine the MPO framework.

Several additional limitations should be noted. First, while docking grid boxes were uniformly set to 25 Å×25 Å×25 Å for all targets, larger boxes (e.g., 30 Å cube) may be warranted to capture peripheral binding interactions for membrane transporters such as PfCRT and PfATP4. Second, protein protonation states were assigned at pH 7.4 without explicit pKa calculations for active-site residues, which may affect hydrogen-bond predictions; PfCRT functions in the acidic digestive vacuole (pH 5.0–5.4), and its protonation state is a specific concern. To quantify this impact, we re-protonated the PfCRT structure at pH 5.2 using pKa predictions from PROPKA3[59] followed by PDBFixer[60] and re-docked the top 100 PfCRT-targeted candidates. Eight active-site residues exhibited shifted protonation states at vacuolar pH (ASP271, pKa 6.27; ASP274, pKa 7.05; GLU49, pKa 7.35; GLU143, pKa 6.53; GLU177, pKa 6.10; HIS68, pKa 6.18; HIS125, pKa 5.78; HIS218, pKa 6.74). The ranking correlation between pH 7.4 and pH 5.2 docking scores was moderate (Spearman $\rho = 0.270$, $p = 0.007$), with a systematic shift of $2.2\,\mathrm{kcal\,mol^{-1}}$ in mean affinity, consistent with fewer charge-assisted interactions under acidic conditions. This indicates that pH correction measurably alters PfCRT rankings and is recommended for future PfCRT-targeted campaigns. Other targets operate at neutral pH and are unaffected. Third, all docking was performed against single static crystal structures without accounting for protein conformational ensembles or resistance mutations (e.g., PfCRT K76T/K76A, PfDHFR N51I/C59R/S108N/I164L). Molecular dynamics simulations would be needed to validate top candidates against these mutations and assess their resilience to these mutations. Fourth, centroid-based sampling may not fully capture activity cliffs within clusters where minor structural changes produce large potency differences; the twofold intra- over inter-cluster Tanimoto separation (mean 0.227 vs 0.101) indicates coherent clustering, but we recommend full-cluster rescoring for top-priority clusters during hit-to-lead optimization. We performed this rescoring for the top-20 MPO clusters (1 815 molecules) and showed limited docking-score dispersion ($\sigma \leq 0.38\,\mathrm{kcal\,mol^{-1}}$) in the tested clusters; this does not exclude unsampled activity cliffs or experimental potency cliffs, while the best member consistently outperformed its centroid (Δ up to $-0.66\,\mathrm{kcal\,mol^{-1}}$); this observation is limited to the tested docking scores and does not establish prospective potency. Fifth, the 19 913 synthesizable leads require experimental IC_{50} testing before any biological conclusions can be drawn. Sixth, polypharmacology predictions (multiple multi-target candidates) are computational hypotheses that require orthogonal experimental confirmation (enzyme assays, cellular phenotypic screens). Seventh, the prospective DEKOIS external validation (PfDHFR, Vina only) achieved ROC-AUC 0.450, near-random enrichment, confirming that AutoDock Vina docking alone cannot discriminate actives from property-matched decoys.[36] The consensus MMV enrichment (ROC-AUC 0.924 to 1.000) incorporates DiffDock ML rescoring, which provides the critical enrichment gain; however, the MMV benchmark remains a positive-control (in-distribution) rather than a fully independent decoy set. The consensus enrichment is also dominated by PfCRT, which accounts for 87.0 % (421/484) of centroid target assignments, whereas only 2.7 % (13/484) of centroids are selected by PfDHFR; the per-target spread in consensus performance (ROC-AUC 0.924 for PfDHFR versus 1.000 for PfATP4) is consistent with ML-based DiffDock rescoring contributing to the observed benchmark separation, although the different benchmark designs do not isolate that contribution. Prospective experimental testing of top-ranked compounds against *Pf* 3D7 and drug-resistant strains is therefore needed to assess the protocol's biological and prospective performance.

Eighth, the PfCRT structure used for docking (PDB: 6UKJ) corresponds to the 3D7 isoform; the chloroquine-resistant 7G8 isoform prevalent in field isolates differs by four key mutations (C72S, K76T, A220S, N326D) that significantly alter the drug-binding pocket geometry. Docking against the 3D7 isoform may not fully recapitulate the binding landscape of the 7G8 variant, and prospective studies should consider isoform-specific docking. Similarly, the PfATP4 docking was performed against an apo-like inward-facing conformation (PDB: 9N10), which may not capture the outward-facing (E2) state critical for ATP-dependent ion efflux. The E1/E2 conformational cycle presents substantially different binding cavities at the membrane-cytosol interface, and our static apo-structure docking cannot account for this conformational plasticity. Membrane-embedded molecular dynamics simulations would also be needed to address these limitations, capture induced-fit stabilization, and account for conformational plasticity of the membrane transporters.

5 Conclusion

This study establishes a resource-efficient computational framework for antimalarial drug discovery by integrating deep generative chemistry with a dual-filter consensus docking and multi-parameter optimization (MPO) approach. By employing variational autoencoder (VAE) clustering for centroid-based sampling and Metropolis-Hastings MCMC for latent space exploration, we substantially reduced virtual screening costs. Prospective DEKOIS PfDHFR docking (Vina only, ROC-AUC 0.450) establishes that structure-based docking alone cannot discriminate actives from property-matched decoys; the higher consensus-score ROC-AUC (0.924 to 1.000) against the MMV Malaria Box positive-control set is consistent with — but does not directly demonstrate, as the consensus was not run on the DEKOIS set — the ML-based DiffDock rescoring providing the enrichment gain. Local exploration of the VAE latent space yielded a modest MPO gain (mean 0.009, best chain 0.025) on a weak surrogate (R^2 0.377). This methodology enables the systematic exploration of massive molecular libraries (e.g., 65 856 molecules) using standard hardware, addressing a fundamental computational bottleneck in the characterization of diverse chemical spaces.

Our results demonstrate the effectiveness of combining African natural product motifs with synthetic drug optimization through hybrid pharmacophore design. The identification of 19 913 leads predicted to be synthesizable (from 53 unique clusters, mean cluster size 763.8) with favorable predicted selectivity and drug-likeness indicates that generative expansion can preserve privileged bioactive scaffolds while substantially diversifying chemical space. The contrast between the DEKOIS Vina-only baseline and the historical MMV positive-control analysis is retained as a calibration of the original workflow, not as proof of prospective accuracy. The corrected target panel adds a four-target, anchor-aware modelling layer and the P2 docking-RRS/polypharmacology analysis adds resistance-oriented prioritization. Together, these components define a transparent computational hypothesis space across DHFR, CRT, ClpP, and ATP4 while preserving the distinction between docking scores, resistance proxies, and biological measurements.

The accessibility of the documented virtual-screening workflow presented here offers a resource-efficient approach to drug discovery in resource-limited settings. By reducing the reliance on massive high-performance computing infrastructure, this framework enables researchers in malaria-endemic regions to systematically prioritize local chemical space for antimalarial candidates. Ultimately, the open-source implementation and computational prioritization of multiple antimalarial leads provide a foundation for collaborative drug development efforts targeting drug-resistant malaria and other neglected tropical diseases.

Supporting information

Supporting information available at the publisher's website. Supplementary Tables S1 to S6, S8 to S10, S12 to S15, S17 to S22, S24 and S27 to S34 and Supplementary Figures S1 to S6 and S8 to S10. MPO equations archived in repository.

Author Contributions

Conceptualization and methodology, M.V.S.T. and S.G.N.E.; software, M.V.S.T. and J.-P.T.N., P.S.; validation, M.V.S.T., J.-P.T.N. and W.F.M.; formal analysis and investigation, M.V.S.T. and S.G.N.E.; resources, S.G.N.E. and W.F.M., P.S.; data curation, M.V.S.T.; writing–original draft preparation, M.V.S.T.; writing–review and editing, M.V.S.T., J.-P.T.N., P.S., W.F.M. and S.G.N.E.; visualization, M.V.S.T.; supervision, S.G.N.E. and W.F.M.; project administration, S.G.N.E. All authors read and agreed to the published version.

Notes

An AI language model (Claude, Anthropic) was used to assist with manuscript formatting, including condensing the abstract and title to meet journal length requirements. All AI-generated suggestions were critically reviewed, modified, and approved by the authors. The scientific content, data analysis, methodology, and interpretation are entirely the work of the authors.

The authors declare no competing financial interest.

Funding

This research received no external funding.

Acknowledgments

We acknowledge the Ersilia Open Source Initiative (https://www.ersilia.io, accessed on December 2025) for machine learning models, ChEMBL and DrugBank for compound data, MMV for Malaria Box, and RDKit.

Data availability

SMILES strings for all 65 856 compounds, primary lead structures, computational workflows, and docking results are available in open formats from the project repository (https://github.com/NanaEngo/Malaria_codesV2). The repository includes the source datasets, analysis scripts, and a supplementary reproducibility table listing input files, row counts, random seeds where applicable, and SHA-256 file identifiers (Supplementary Table S38).

Scope and limitations. This is a modelling-only study. It provides computational evidence for generative novelty, scaffold recovery, historical positive-control calibration, target-anchored Vina pose generation, docking-derived RRS, and PNS/ACSI prioritization. None of these outputs is an experimental affinity, activity, MD-RRS, or biological validation result. The target-panel matrix is retained as raw computational evidence with its target-specific provenance; the legacy DiffDock consensus and the superseded four-target consensus are excluded. Additional limitations include the 3D7 PfCRT isoform (PDB 6UKJ), the Y01 membrane-proxy anchor, the motif-defined PfATP4 region in an apo inward-facing structure (PDB 9N10), rigid-receptor docking, static mutation models, and the absence of experimental *Plasmodium* testing. These boundaries define the intended scope of the paper rather than a deficiency hidden by the computational analysis.

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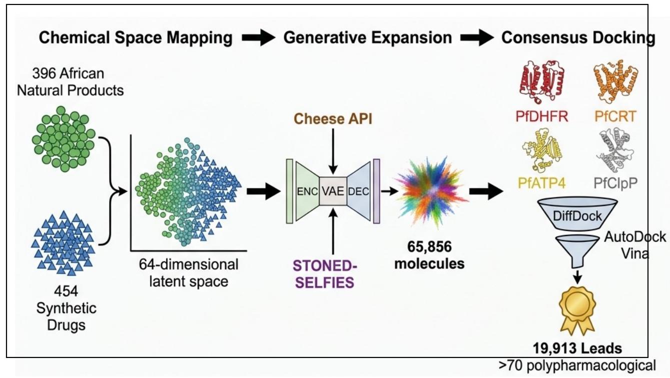
TOC Graphic


Chemical Space Mapping
Generative Expansion
Consensus Docking
396 African Natural Products
454 Synthetic Drugs
64-dimensional latent space
Cheese API
ENC
VAE
DEC
STONED-SELFIES
65,856 molecules
PfDHFR
PfCRT
PfATP4
PfClpP
DiffDock
AutoDock Vina
19,913 Leads
>70 polypharmacological